

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 13)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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8	
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11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Explain why **proxy data** are needed to study the early conditions on Earth.

[1 mark]

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0 1 . 2

Complete Table 1 to show what each analytical technique can be used to estimate.

[4 marks]

Table 1

Technique	What it estimates	
Ratio of carbon-12 to carbon-14 in a sample		
Ratio of oxygen-18 to oxygen-16		
Analysis of air bubbles trapped in ice cores		
Ratio of magnesium to calcium in calcite deposits		

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0 1 . 3

Explain why an understanding of continental drift and ocean currents has been important in explaining past changes in conditions.

[2 marks]

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01 . 4

Life probably first developed around volcanic vents on the seabed. Explain why the conditions in the oceans were more suitable than those on land 3.5 billion years ago.

[3 marks]

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02 . 1

State what is meant by **hybrid vigour**.

[1 mark]

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02 . 2

Explain how cross-breeding Zebu and Ayrshire cattle produced a breed suited to hot climates.

[2 marks]

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02 . 3

Give **two** potential applications of cloning to livestock farming.

[2 marks]

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03

Table 1 shows the yield of a vegetable crop under different pest control methods over three years on one farm.

Table 1

Method	Mean yield / t ha ⁻¹	Pesticide applications per year	Non-target insect species recorded in field margins
Chemical insecticide only	18.5	6	8
Crop rotation + beetle banks + hedgerows	16.2	0	31
Integrated control	18.1	1-2	26

03**1**

Explain how crop rotation and beetle banks reduce pest damage.

[4 marks]

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03**2**

Use Table 1 to evaluate integrated control compared with the other two methods.

[3 marks]

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Describe how the sterile male technique controls insect pests and explain why it is species-specific.

[3 marks]

10

0	4
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Table 1 shows features of some fishing methods.

0	4	.	1
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Complete Table 1.

[6 marks]

Table 1

Method	Target species	One environmental disadvantage	
Demersal trawling			
Purse seining			
Pelagic long-lining			

0 4 . 2

Explain why by-catch is a particular problem with shrimp trawling and drift netting.

[2 marks]

0 4 . 3

Explain what is meant by **ghost fishing** and describe **two** ways it can be reduced.

[2 marks]

10

0 5 . 1

Explain how each of the following environmental impacts of salmon farming arises and how it can be reduced.

[8 marks]

Overfishing of wild fish for feed

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Sea lice and disease

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Escapes affecting wild gene pools

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Organic waste pollution

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05 . 2 Explain why culling seals and cormorants near fish farms is controversial and give an alternative.

[2 marks]

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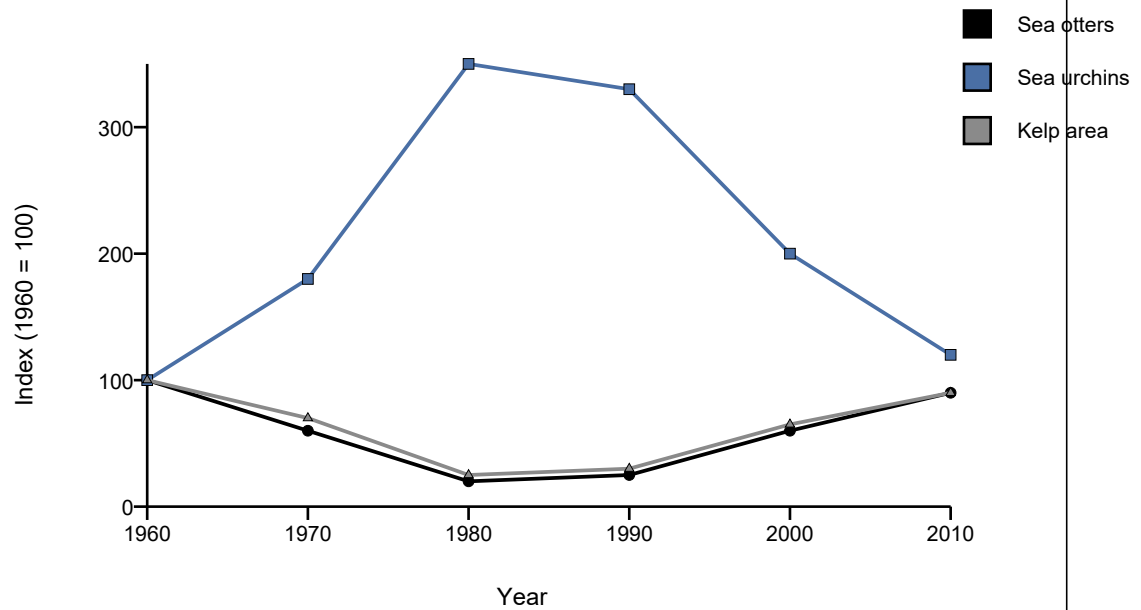
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10

0 6

Figure 1 shows the estimated populations of sea otters, sea urchins and the area of kelp forest along part of the coast of California between 1960 and 2010.

Figure 1



0 6

1 Describe the relationship between the three variables shown in Figure 1.

[3 marks]

0 6

2 Explain the relationships shown in Figure 1.

[3 marks]

06 . 3

Give **two** other examples of human over-exploitation of one species affecting the population of another.

[2 marks]

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2

06 . 4

Explain why the loss of beavers or forest elephants can cause the decline of many other species.

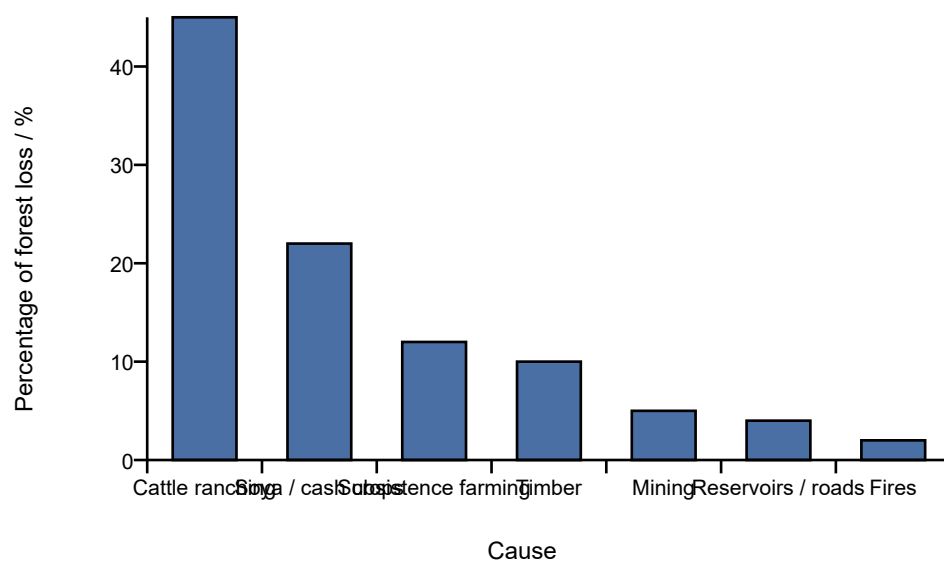
[2 marks]

10

07

Figure 1 shows the main causes of tropical rainforest loss in a South American country between 2000 and 2020.

Figure 1



07 . 1

Explain why traditional 'slash and burn' subsistence agriculture was sustainable but has become a threat.

[3 marks]

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07 . 2

Explain why clearance for cattle ranching or cash crops is usually unsustainable.

[2 marks]

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07 . 3

Describe how each of the following threatens rainforest.

[3 marks]

Mineral extraction

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HEP reservoirs

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Climate change

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07 . 4

Describe how a debt-for-nature swap works.

[2 marks]

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08 . 1

Explain why food surpluses produced in MEDCs in the 1970s could not easily be sold.

[3 marks]

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08 . 2

Describe **four** methods used to reduce over-production while maintaining farm incomes.

[4 marks]

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08 . 3

Describe **three** features that farmers may be paid for under an agri-environmental scheme and the conservation purpose of each.

[3 marks]

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09 Table 1 compares two exploited species.

Table 1

Feature	Anchovy	Blue whale
Age at sexual maturity	1 year	10 years
Offspring per female per year	Thousands of eggs	1 calf every 2-3 years
Lifespan	3-4 years	80+ years
Parental care	None	Extensive

09 . **1** Use Table 1 to identify which species is r-selected and which is k-selected, and explain the difference between the strategies.

[3 marks]

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09 . **2** Explain why k-selected species are much more easily over-exploited than r-selected species.

[3 marks]

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09 . 3

State the equation used to forecast a change in population size.

[1 mark]

09 . 4

Explain what is meant by **maximum sustainable yield** and how it relates to carrying capacity.

[3 marks]

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10 . 1

Evaluate the relative importance of ex-situ and in-situ methods in the conservation of endangered species.

[9 marks]

[illegible]

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10 . **2** State what is meant by **rewilding**.

[1 mark]

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10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.

[25 marks]

OR

$$\boxed{1} \boxed{1} . \boxed{2}$$

Discuss the extent to which the principles of the circular economy could be used to develop sustainable lifestyles.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

Do not write outside the box

END OF QUESTIONS